

ASSIGNMENT 6

Instructions for Students

- Create a **separate HTML and CSS file** for each question.
 - **Do not** use JavaScript or any external CSS frameworks.
 - You may use semantic HTML5 tags (**header, main, section, footer**, etc.).
 - All designs should be **responsive** and use **only Flexbox or Grid** for layout.
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1. Create a **personal portfolio webpage layout** using **Flexbox**.

Requirements:

- Header: includes your name (left) and navigation links (right).
- Main area: two sections — an “About Me” section (text) and an image section.
- Footer: simple copyright text centered.
- Use **display: flex;** to align header items horizontally and main content side-by-side.
- Ensure that when the screen width is below 700px, all sections stack vertically.
- Center items both vertically and horizontally using **align-items** and **justify-content**.

2. Design a **4-column product gallery** using **CSS Grid**.

Requirements:

- Create a grid of 8 product cards.
- Each card should contain a product name and price.
- Use **grid-template-columns: repeat(4, 1fr);** for desktop.
- Adjust to 2 columns on tablets and 1 column on mobile using media queries.
- Add hover effects on each card: use **transform: scale(1.05)** and **transition: 0.3s**.

- Cards should have equal height regardless of content length.

3. Create a **3-section homepage** using **Flexbox**: Header, Main, and Footer.

Requirements:

- Header: navigation links spaced evenly using **justify-content: space-around;**
- Main: three boxes (Services, Projects, Contact) horizontally aligned.
- On hover, each box should slightly rotate (**transform: rotate(2deg);**).
- Footer: stays at the bottom and spans full width.
- The layout must remain balanced at all screen sizes.

4. Build a **dashboard layout** using **CSS Grid**.

Requirements:

- Header spans the top across all columns.
- Sidebar on the left with vertical navigation.
- Main content with 4 boxes arranged in 2 rows and 2 columns.
- One box should span two columns (**grid-column: span 2;**).
- Use **grid-gap** for spacing and fractional units (**fr**) for widths.
- Add hover effects that scale up the box and cast a shadow.

5. Design a **pricing section** using **Flexbox**.

Requirements:

- Create three pricing cards aligned horizontally.
- The middle card should be slightly larger and highlighted with a different background color.
- On hover, cards should rise up slightly (**transform: translateY(-10px)**), with a smooth transition.

- Add a simple fade-in animation for all three cards when the page loads using **@keyframes**.
- Make the layout stack vertically below 750px.

6. Create a **responsive blog layout** using **CSS Grid**.

Requirements:

- Header with title and nav links.
- Sidebar on the left containing categories.
- Main content area displaying two blog posts side-by-side.
- Footer with contact info.
- Use **grid-template-areas** to structure the layout.
- Below 800px, sidebar should move below the main content.
- Add padding, margins, and equal spacing using **gap**.

7. Design a **CSS loading animation** using only **div** elements.

Requirements:

- Create three small circles that move up and down repeatedly.
- Use **@keyframes** to control the motion and timing.
- Each circle should start its animation with a slight delay.
- Center the animation on the page using Flexbox alignment.
- Use **transform: translateY()** and **animation-iteration-count: infinite;**

8. Build a **photo gallery** using **Grid** with hover transformations.

Requirements:

- Use **grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));**

- Each image should fill its grid cell completely and scale up slightly on hover.
- Add **transition: transform 0.5s ease;** for smooth zoom effect.
- Include image captions below each image using **<figcaption>**.
- Layout should automatically adjust to screen size.

9. Create a **card flip animation** using **CSS Transformations and Transitions**.

Requirements:

- Build one card that flips 180 degrees on hover.
- Front side: shows title and image.
- Back side: shows text description.
- Use **transform-style: preserve-3d;** and **perspective: 1000px;**
- Implement the flip using **transform: rotateY(180deg);** and smooth transitions.
- Card must remain centered on screen using Flexbox.

10. Design a **responsive webpage layout** combining **Flexbox and Grid**.

Requirements:

- Header with logo and nav links using Flexbox.
- Main area divided into a **2-column Grid** (Article + Sidebar).
- Footer spans the full width with centered text.
- Inside the main article, include three boxes aligned using Flexbox.
- Use **@keyframes** to animate one element (e.g., a title fades in or moves up).
- Ensure the layout collapses into a single column on mobile screens.